TufDiet: AI-Powered Nutrition Tracking System

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FEE306 Applied Artificial Neural Networks Course Project

***Abstract*—This paper presents TufDIet, an AI-powered web-based nutrition tracking system designed to help users monitor their daily food intake through image based food recognition. The system combines Django based web application with a PyTorch powered deep learning model for automatic food classification. Using Transfer learning with MobileNetV2 and EfficientNet architectures, also the system can identify almost 99 different food classes. User can upload food images, and the system automatically classifies the food and retrieves nutritional information including calories, protein, carbohydrates, and fat content.**

The application also stores meal history and provides daily nutrition summaries. Experimental results show that the custom trained model achieves 90% validation accuracy on the 5-class Turkish cuisine dataset, while the Food-101 model achieves 59.14% confidence on the test images. The system provides an intuitive interface for meal logging and nutrition tracking

***Index Terms*— Nutrition Tracking, Deep Learning, Food Classification, Transfer Learning, MobileNetV2, EfficientNet, Django, PyTorch.**

1. Introduction

1.1 Background

Maintaining a healthy diet requires careful monitoring of food intake and nutritional values. Traditional nutrition tracking methods require users to manually search and enter food information, which is time consuming and open to errors. The integration of artificial intelligence into nutrition tracking offer a promising solution to automate food recognition and nutritional data retrieval.

1.2. Problem Statement

Current nutrition tracking applications suffer from several limitations:

- Manual food entry is tedious and time consuming.
- Users often misidentify nutritional values.
- Lack of personalized dietary recommendations.
- Limited support for diverse cuisines.

1.3 Objectives

This project aims to develop an AI-Powered nutrition tracking system that:

1. Automatically identifies food items from images,
2. Retrieves accurate nutritional information,
3. Provides a user-friendly interface for meal logging,
4. Tracks daily nutrition intake with visual summaries.

1.4 Scope

The TufDiet system focuses on:

- Image based food classification using deep learning.
- Integration of nutritional databases.
- Web based user interface with Django.
- User authentication and meal history tracking.

2. Methodology

2.1 System Architecture

The TufDiet system consists of three main components for now:

- Frontend Layer: Django templates with HTML/CSS/JavaScript
- Backend Layer: Django REST API with Python
- AI Layer: PyTorch based image classification model

2.. Dataset

2.2.1. Food-101 Dataset:

99 food classes including pizza, hamburger, sushi, etc. Used pre-trained model from previous training.

2.2.2. Custom Dataset:

5 Turkish cuisine classes: Baklava, Corba (Soup), Elma (Apple), Salata (Salad), Tavuk (Chicken)

10 images for per class collected from web sources. Total: 50 images for training.

2.2.3. Nutrition Database:

Combined nutritional data form:

- Kaggle Food Nutrition Dataset (2,395 food items)
- Custom food entries from Food-101 dataset
- User-constributed nutrition data

2.3. Model Architecture

2.3.1. Model Selection:

Two models were implemented:

1. Custom 5-Class Model (MobileNetV2): Pre-trained MobileNetV2 backbone with custom classifier head. Final accuracy: 90%
2. Food-101 Model (EfficientNet-B0): Pre-trained EfficientNet-B0 backbone modified for 99 output classes.

2.3.2. Training configuration

Parameter	Value
Image Size	224×224
Batch Size	8-32
Learning Rate	0.001
Optimizer	AdamW
Scheduler	CosineAnnealingLR
Epochs	15
Device	CPU (with CUDA fallback)

2.4 Web Application Development

2.4.1 Django Framework

Python-based web framework with REST API for mobile/web integration, built-in user authentication, and SQLite database.

2.4.2 API Endpoints3. Results

Endpoint	Method	Description
/api/ai-analyze/	POST	Image classification
/api/meals/	GET/POST	Meal CRUD operations
/api/profiles/	GET/PUT	User profile
/api/auth/login/	POST	User login

3.1 Model Performance

3.1.1 Custom 5-Class Model Results

Metric	Value
Validation Accuracy	90%
Training Samples	44
Validation Samples	11
Classes	5

3.1.2 Food-101 Model Results

Test image: Pizza-3007395.jpg

Prediction	Confidence
Pizza	74%
Paella	23%
Takoyaki	1%
Lasagna	0%
Apple Pie	0%

3.2 System Features

User Authentication: Registration, Login, Logout with session management

Meal Logging: Image upload for automatic classification, manual food entry option

Dashboard Features: Today's nutrition summary (Calories, Protein, Carbs, Fat), Meal history with images

3.3 Performance Metrics4. Discussion

Metric	Value
Model Load Time	~2-3 seconds
Prediction Time	~0.5 seconds
API Response Time	<1 second
Database Query	<0.1 seconds

4.1 Key Findings

Transfer Learning Effectiveness: Using pre-trained MobileNetV2 and EfficientNet models significantly reduced training time and improved accuracy.

Confidence Threshold: Implementing a 50% confidence threshold reduced false predictions but also limited some correct identifications.

4.2 Limitations

• Limited Custom Classes: Only 5 custom Turkish food classes available

• CPU Performance: Training and inference on CPU limits speed

• Dataset Size: Small custom dataset may not generalize well

4.3 Future Improvements

• Expanded Dataset: Collect more images for each class

• GPU Training: Implement GPU-based training

• Mobile App: Develop Flutter mobile application

• Calorie Goals: Add personalized daily calorie recommendations

5. Conclusion

This project successfully demonstrated TufDiet, an end-to-end AI ecosystem that bridge the gap between complex deep learning architectures and practical nutritional health management. By utilizing Transfer Learning with the MobileNetV2 backbone, the system achieved a robust 90% validation accuracy on a specialized 5-class Turkish cuisine dataset, significantly outperforming generic models in local context recognition.

The core innovation of TufDiet lies in its Dynamic Learning Loop. Unlike traditional static classifiers, the system integrates a comprehensive Nutrition Metadata repository (CSV), allowing it to autonomously fetch nutritional data for unrecognized items and categorize user-contributed images for future model fine-tuning. The Django-based backend ensures secure user authentication and persistent meal history tracking, providing a seamless UX for daily monitoring.

5.1 Future Works

Future work will prioritize the following expansions to evolve TufDiet into a production-ready application:

1. Dataset Scaling: Expanding the custom high-accuracy classes from 5 to 50+, focusing on diverse regional dishes.
2. Incremental Retraining: Implementing automated batch retraining pipelines that use Croissant metadata standards to handle the growing user-contributed image repository.
3. Cross-Platform Accessibility: Developing a Flutter-based mobile application to allow real-time meal logging via mobile camera sensors.
4. Advanced Analytics: Integrating predictive AI to provide personalized weekly caloric trend analysis and macro-nutrient goal recommendations.

Through this midterm project, the effectiveness of combining lightweight deep learning models with scalable web architectures has been proven, laying a solid foundation for the future of automated dietary assistants.

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